

# USING MACHINE LEARNING TO ACCELERATE THE SOLUTION OF LINEAR EQUATIONS

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**Abstract.** This work presents the integration of a Machine Learning (ML) approach into Computational Fluid Dynamics (CFD), utilizing a K-Nearest Neighbors (KNN) model to provide an initial guess for each time step of the simulation. The method is tested on the unsteady, 2D heat diffusion equation. Comparisons between the traditional CFD solution and the hybrid approach show that the mean absolute error between the two methods remains below 1.0%, while reducing computational costs by 60%. Despite the simplicity of the chosen model equation, the results demonstrate that significant computational savings can be achieved even with a basic KNN-based ML model.

**Key words:** *KNN; CFD; machine learning; linear regression; hybrid approach; computational time*

## 1 Introduction

In the process of CFD solving, traditional numerical methods like finite difference, finite volume, finite element, and spectral methods achieve high accuracy but are computationally expensive for real-time applications due to their high degrees of freedom. Brunton [1] demonstrated the potential of combining machine learning with CFD, highlighting that the integration of machine learning can enhance the understanding and control of complex flow problems. Kochkov [2] validated his conclusions through simulation experiments. Vinuesa [3] also pointed out that machine learning can significantly reduce the computational cost of CFD simulations. The integration of machine learning with CFD can greatly improve both the efficiency and accuracy of simulations. By using deep learning to model the spatial derivatives in finite volume methods or finite difference methods, it is possible to significantly reduce the required grid resolution and computation time without sacrificing simulation accuracy. The hybrid simulation framework proposed by Wen [4], which combines deep learning with CFD simulations, effectively reduces error accumulation, maintaining the accuracy of physical fields in long-term simulations and significantly saving computation time. However, the time cost of training the models cannot be overlooked. Muther [5] demonstrated that by embedding physical laws into machine learning models, it is possible to generate predictions that adhere to physical principles even with limited data.

Therefore, this work presents the implementation of a machine learning (ML) approach into computational fluid dynamics (CFD) for implicit time integration schemes in the heat diffusion equation. While the selected equation is a simple model equation, this approach shows that substantial computational savings are possible even with a simple KNN-based ML model. Further studies on non-linear equations and more sophisticated ML models and architectures are planned for the future.

## 2 Problem description

This section introduces a methodology that combines the K-nearest neighbours (KNN) algorithm with the unsteady, two-dimensional heat diffusion equation. The approach begins with the solution of the

unsteady heat diffusion equation, which is described by the following partial differential equation (PDE):

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T \quad (1)$$

where  $T$  is the temperature,  $t$  is time,  $\alpha$  is the thermal diffusivity, and  $\nabla^2 T$  represents the Laplacian operator, which involves spatial second derivatives. This equation models describes the time-dependent heat diffusion in a material and serves as the foundation of numerical simulations.

The equation is discretized in time and space for numerical solutions, with the time derivative approximated by an implicit first-order method and the spatial derivatives by a second-order finite difference scheme, leading to the discretized form:

$$\frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} = \alpha \left[ \frac{T_{i+1,j}^{n+1} - 2T_{i,j}^{n+1} + T_{i-1,j}^{n+1}}{\Delta x^2} + \frac{T_{i,j+1}^{n+1} - 2T_{i,j}^{n+1} + T_{i,j-1}^{n+1}}{\Delta y^2} \right] \quad (2)$$

where  $\Delta t$  is the time step, and  $\Delta x$  and  $\Delta y$  are the spatial step sizes in the x- and y-directions, respectively.

Now we introduce the following coefficients

$$\begin{aligned} a_P &= \left[ \frac{1}{\Delta t} + \frac{2\alpha}{\Delta x^2} + \frac{2\alpha}{\Delta y^2} \right] \\ a_E &= a_W = \left[ -\frac{\alpha}{\Delta x^2} \right] \\ a_N &= a_S = \left[ -\frac{\alpha}{\Delta y^2} \right] \\ b &= \frac{T_{i,j}^n}{\Delta t} \end{aligned}$$

Which leads to the following, simplified equation:

$$a_P T_{i,j}^{n+1} + \sum_k^{N,E,S,W} a_k T_k^{n+1} = b \quad (3)$$

This can now be solved for  $T_{i,j}^{n+1}$ , and using this equation is known as the Gauss-Seidel algorithm:

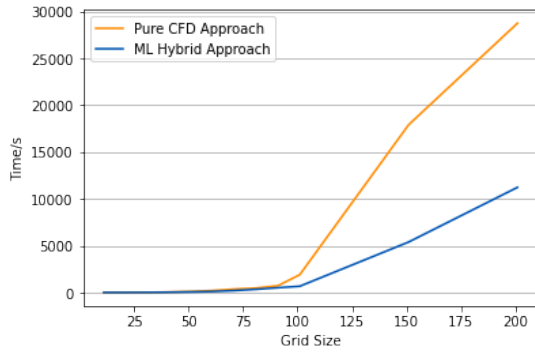
$$T_{i,j}^{n+1} = \frac{1}{a_P} \left( b - \sum_k^{N,E,S,W} a_k T_k^{n+1} \right) \quad (4)$$

Machine learning models, particularly K-Nearest Neighbors (KNN), are employed to predict the initial temperature values in the linear regression equation, improved starting points for the Gauss-Seidel iterative solver and enhancing computational efficiency. The solution of the linear system  $\mathbf{Ax} = \mathbf{b}$  in implicit time integration schemes requires an initial guess for  $\mathbf{x}$ , often based on initial conditions or previous iterations. More accurate initial guesses can reduce computational costs. As a non-parametric method, KNN predicts the temperature at each point by evaluating the values of its  $k$  nearest neighbours, effectively capturing non-linear relationships between the current and previous temperature distributions. Trained on a dataset of heat diffusion equation solution under various boundary conditions, the KNN model can estimate  $T_{i,j}^{n+1}$  by interpolating from the closest training samples.

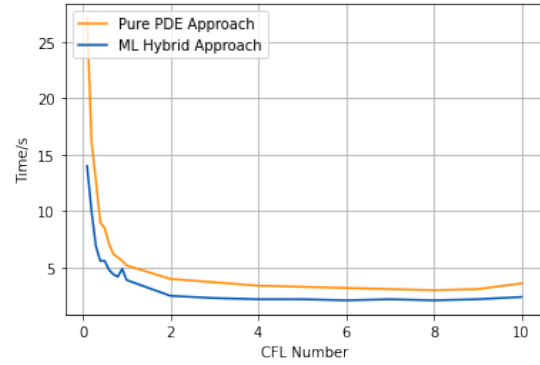
Training involves solving the heat diffusion equation  $X$  times, each with a distinct set of boundary conditions, where  $X$  represents the total number of simulations performed. The training data is generated using grids of varying sizes, with the smallest grid containing  $10 \times 10$  cells and the largest being  $200 \times 200$  cells. In this study,  $X=6$  simulations are performed, the smallest grid yields 600 training points ( $10 \times 10 \times 6$ ), while the largest grid yields 240,000 training points ( $200 \times 200 \times 6$ ). After sufficient training, the model is tested on new simulations with boundary conditions not seen in training.

### 3 Numerical results

In the following, a hybrid approach integrating machine learning methods into computational fluid dynamics is compared against the traditional finite element method for solving heat diffusion equations. By evaluating computation times across different grid sizes and CFL numbers, the advantages of the hybrid approach are demonstrated.



(a) A comparative analysis of the time required to solve the heat diffusion equation using the pure CFD approach and the ML hybrid approach under different grid sizes is presented in the following graph.



(b) A comparative analysis of the time required to solve the heat diffusion equation using the pure CFD approach and the ML hybrid approach under different CFL number is presented in the following graph.

Figure 1: A comparative analysis of the time required to solve the heat diffusion equation using the pure CFD approach and the ML hybrid approach under different grid sizes and CFL number is presented in the following graph.

Figure 1 presents the time required for the pure CFD approach and the hybrid ML approach to solve the heat diffusion equation under the same boundary conditions, with varying grid sizes and CFL numbers. Figure 1a shows a comparison of the simulation time for both methods at a CFL value of 10, for different grid sizes. As the grid size increases, the advantages of the ML hybrid approach become more pronounced. Compared to the pure CFD approach, the hybrid method with KNN requires only 40% of the simulation time. Figure 1b presents a comparison of the simulation time for both methods at a grid size of 20, for different CFL numbers. Upon observing the results for various CFL values, it is found that the CFL value does not significantly affect the simulation time. Regardless of the CFL value, the hybrid method consistently saves approximately half of the computation time. The hybrid approach demonstrates superior computational speed compared to the traditional solver, while maintaining

|                    | $10 \times 10$ | $100 \times 100$ |
|--------------------|----------------|------------------|
| Pure CFD Approach  | 0.3s           | 17892.7s         |
| ML Hybrid Approach | 0.2s           | 5385.4s          |
| Time Saving        | 33.3%          | 69.9%            |
| MAE                | 0.02%          | 1.0%             |

Table 1: Comparison of time and mean absolute error(MAE) with two approach at grid size= $10 \times 10$  and  $100 \times 100$

comparable accuracy, with the mean absolute error (MAE) consistently remaining below 1.0%. Table 1 presents a comparison of the time required by both methods to solve the same case. For a grid size of  $10 \times 10$ , the ML hybrid approach reduces computation time by 33.3%, with an MAE of 0.02%. In the case of a  $100 \times 100$  grid, the ML hybrid approach results in a 69.9% time savings, with an MAE of 1.0%. These results further confirm that the ML hybrid approach not only ensures simulation accuracy but also significantly enhances computational efficiency.

#### 4 Conclusions

In this research, the machine learning hybrid approach has been utilized to predict the results of the linear regression equation. This method was applied to the 2D heat diffusion equation and tested through the CFL number and grid size. The results show that, compared to the pure CFD approach, the outcomes obtained using the ML hybrid approach are nearly identical, with the mean absolute error between the two approaches remaining below 1.0%, thereby validating the effectiveness of the results in this test case. Furthermore, the ML hybrid approach significantly improves computational efficiency, with the performance enhancement becoming more pronounced as the grid size increases, achieving a time saving of up to around 60%. The ML hybrid approach proves to be an effective method for improving the computational efficiency of CFD simulations, which aligns with the findings in the literature [4]. To obtain more generalized results, the next step is to extend the hybrid approach to the non-linear regression equation to assess both computational savings and accuracy.

#### REFERENCES

- [1] Brunton, S.L. and Noack, B.R. *Machine learning for fluid mechanics. Annual review of fluid mechanics*, **52**(1):477–508, (2020).
- [2] Kochkov, D. and Smith, J. A. *Machine learning–accelerated computational fluid dynamics. Proceedings of the National Academy of Sciences*, **118**(7):e2101784118, (2021).
- [3] Vinuesa, Ricardo and Brunton, Steven L. *Enhancing computational fluid dynamics with machine learning. Nature Computational Science*, **2**(6):358–366, (2022).
- [4] Kaijie Wen and Li Guo. *A hybrid simulation method integrating CFD and deep learning for gas–liquid bubbly flow. Chemical Engineering Journal*, **495**:153–515, (2024).
- [5] Muther, Temoor and Dahaghi, Amirmasoud Kalantari. *Physical Laws Meet Machine Intelligence: Current Developments and Future Directions. Artificial Intelligence Review*, **56**(7):6947–7013, (2023).